ETALE LOCALITY ON THE SOURCE PLUS SMOOTH LOCALITY ON THE TARGET IMPLIES LOCALITY ON THE SOURCE-AND-TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's own remark that the naive translation-invariance argument would require extra etale localization on the target and would become annoying, followed by the choice to use the automorphisms of the group action lemma instead; the remark is reproduced verbatim and is part of the argument's structure.

Editorial note. The proof also contains the parenthetical hint '(Details omitted; hint: compute the derivatives.)' for the etaleness of the p-th power map; the parenthetical is reproduced verbatim and the surrounding argument is complete as written.

Editorial note. The counted spans carry 5 navigation stubs; one of them points at the published item of this chapter whose label is lemma-orbits, which is used here as a prerequisite rather than restated, so this card is not a duplicate of it.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Morphisms of germs attached to a property and the locus on which the property holds. Same chapter, Lemma lemma-largest-open-of-the-source, named in the statement at line 7931. Supplies the object W(-) being compared.
- Smooth morphisms are etale locally standard smooth over an affine space. Morphisms, Lemma morphisms-lemma-smooth-etale-over-affine-space, invoked at proof lines 7950-7952. Supplies the etale morphism of the target to an affine space over the base.
- Etale locality on the source implies etale locality on the source-and-target, and the comparison of loci along the resulting square. Same chapter, Lemmas lemma-etale-local-source-target and lemma-etale-etale-local-source-target, invoked at proof lines 7961-7966 and 8033. Supplies the identification of the loci and the invariance statement used with the Frobenius-like map.
- The group of elementary automorphisms of affine space and the structure of its invariant open subschemes. Same chapter, Lemma lemma-orbits,

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invoked at proof lines 7981 and 8006-8011. Supplies the description of the invariant open in each fibre, which is then excluded in characteristic p.

Source reference. Original lemma 0CEZ: sources/stacks-descent/pinned-original-source.tex, source lines 7787–7787. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 04R2: sources/stacks-descent/pinned-original-source.tex, source lines 7532–7532. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0CEY: sources/stacks-descent/pinned-original-source.tex, source lines 6890–6890. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0CF0: sources/stacks-descent/pinned-original-source.tex, source lines 7847–7847. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 054L: sources/stacks-chow/tags.txt, source lines 6644–6644. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $\mathcal{P}$ be a property of morphisms of schemes. Assume*

(1) *$\mathcal{P}$ is étale local on the source,*
(2) *$\mathcal{P}$ is smooth local on the target,*
(3) *$\mathcal{P}$ is stable under postcomposing with open immersions: if $f : X \to Y$ has $\mathcal{P}$ and $Y \subset Z$ is an open subscheme then $X \to Z$ has $\mathcal{P}$.*

Given a commutative diagram of schemes

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g} \\ X & \xrightarrow{f} & Y \end{array} \quad \text{with points} \quad \begin{array}{ccc} x' & \longrightarrow & y' \\ \downarrow & & \downarrow \\ x & \longrightarrow & y \end{array}$$

such that g is smooth y' and $X' \to X \times_Y Y'$ is étale at x', then $x \in W(f) \Leftrightarrow x' \in W(f')$ where $W(-)$ is as in Lemma 0CEY.

Proof. Since $\mathcal{P}$ is étale local on the source we see that $x \in W(f)$ if and only if the image of x in $X \times_Y Y'$ is in $W(X \times_Y Y' \to Y')$. Hence we may assume the diagram in the lemma is cartesian.

Assume $x \in W(f)$. Since $\mathcal{P}$ is smooth local on the target we see that $(g')^{-1}W(f) = W(f) \times_Y Y' \to Y'$ has $\mathcal{P}$. Hence $(g')^{-1}W(f) \subset W(f')$. We conclude $x' \in W(f')$.

Assume $x' \in W(f')$. For any open neighbourhood $V' \subset Y'$ of y' we may replace Y' by V' and X' by $U' = (f')^{-1}V'$ because $V' \to Y'$ is smooth and hence the base change $W(f') \cap U' \to V'$ of $W(f') \to Y'$ has property $\mathcal{P}$. Thus we may assume

there exists an étale morphism $Y' \to \mathbf{A}^n_Y$ over Y, see Morphisms, Lemma 054L. Picture

$$\begin{array}{ccc} X' & \longrightarrow & Y' \\ \downarrow & & \downarrow \\ \mathbf{A}^n_X & \xrightarrow[f_n]{} & \mathbf{A}^n_Y \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

By Lemma 04R2 (and because étale coverings are smooth coverings) we see that $\mathcal{P}$ is étale local on the source-and-target. By Lemma 0CEZ we see that $W(f')$ is the inverse image of the open $W(f_n) \subset \mathbf{A}^n_X$. In particular $W(f_n)$ contains a point lying over x. After replacing X by the image of $W(f_n)$ (which is open) we may assume $W(f_n) \to X$ is surjective. Claim: $W(f_n) = \mathbf{A}^n_X$. The claim implies f has $\mathcal{P}$ as $\mathcal{P}$ is local in the smooth topology and $\{\mathbf{A}^n_Y \to Y\}$ is a smooth covering.

Essentially, the claim follows as $W(f_n) \subset \mathbf{A}^n_X$ is a "translation invariant" open which meets every fibre of $\mathbf{A}^n_X \to X$. However, to produce an argument along these lines one has to do étale localization on Y to produce enough translations and it becomes a bit annoying. Instead we use the automorphisms of Lemma 0CF0 and étale morphisms of affine spaces. We may assume $n \geq 2$. Namely, if $n = 0$, then we are done. If $n = 1$, then we consider the diagram

$$\begin{array}{ccc} \mathbf{A}^2_X & \xrightarrow[f_2]{} & \mathbf{A}^2_Y \\ {\scriptstyle p}\downarrow & & \downarrow \\ \mathbf{A}^1_X & \xrightarrow{f_1} & \mathbf{A}^1_Y \end{array}$$

We have $p^{-1}(W(f_1)) \subset W(f_2)$ (see first paragraph of the proof). Thus $W(f_2) \to X$ is still surjective and we may work with f_2. Assume $n \geq 2$.

For any $1 \leq i, j \leq n$ with $i \neq j$ and $d \geq 0$ denote $T_{i,j,d}$ the automorphism of $\mathbf{A}^n$ defined in Lemma 0CF0. Then we get a commutative diagram

$$\begin{array}{ccc} \mathbf{A}^n_X & \xrightarrow[f_n]{} & \mathbf{A}^n_Y \\ {\scriptstyle T_{i,j,d}}\downarrow & & \downarrow{\scriptstyle T_{i,j,d}} \\ \mathbf{A}^n_X & \xrightarrow{f_n} & \mathbf{A}^n_Y \end{array}$$

whose vertical arrows are isomorphisms. We conclude that $T_{i,j,d}(W(f_n)) = W(f_n)$. Applying Lemma 0CF0 we conclude for any $x \in X$ the fibre $W(f_n)_x \subset \mathbf{A}^n_x$ is either $\mathbf{A}^n_x$ (this is what we want) or $\kappa(x)$ has characteristic $p > 0$ and $W(f_n)_x$ is the complement of a finite set $Z_x \subset \mathbf{A}^n_x$ of closed points. The second possibility cannot occur. Namely, consider the morphism $T_p : \mathbf{A}^n \to \mathbf{A}^n$ given by

$$(x_1, \ldots, x_n) \mapsto (x_1 - x_1^p, \ldots, x_n - x_n^p)$$

As above we get a commutative diagram

$$\begin{array}{ccc} \mathbf{A}^n_X & \xrightarrow{f_n} & \mathbf{A}^n_Y \\ {\scriptstyle T_p}\downarrow & & \downarrow{\scriptstyle T_p} \\ \mathbf{A}^n_X & \xrightarrow{f_n} & \mathbf{A}^n_Y \end{array}$$

The morphism $T_p : \mathbf{A}^n_X \to \mathbf{A}^n_X$ is étale at every point lying over x and the morphism $T_p : \mathbf{A}^n_Y \to \mathbf{A}^n_Y$ is étale at every point lying over the image of x in Y. (Details omitted; hint: compute the derivatives.) We conclude that

$$T_p^{-1}(W) \cap \mathbf{A}^n_x = W \cap \mathbf{A}^n_x$$

by Lemma 0CEZ (we've already seen $\mathcal{P}$ is étale local on the source-and-target). Since $T_p : \mathbf{A}^n_x \to \mathbf{A}^n_x$ is finite étale of degree $p^n > 1$ we see that if Z_x is not empty then it contains $T_p^{-1}(Z_x)$ which is bigger. This contradiction finishes the proof. $\square$